Angel Ivan Rodríguez León

M. Sc. Physics and Chemistry PhD candidate

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Personal Profile

I'm doing my Ph.D. in Chemistry at Northwestern University, where I focus on computational simulations. I've got a background in electronic structure calculations and molecular dynamics simulations. I work with both Prof. Monica Olvera de la Cruz at Northwestern and Prof. Maria K. Chan at Argonne National Lab. I'm especially excited about AI for Science and want to focus on using machine learning to transform how we discover, understand, and design new materials and molecules.

Education

Northwestern University — *Illinois, US*

Ph.D candidate (Chemistry) — *Sep 2024 - present*

- Fulbright Scholar (2024-2027)
- Visiting Graduate Student- NST Division at Argonne National Lab. (2025-2026)
- Advisor: Monica Olvera de la Cruz
- Co-Advisor: Maria K. Chan (Center for Nanoscale Materials-Argonne National Laboratory)

Universidad Nacional Autonoma de Mexico — *CDMX, MX*

Master of Science (Physics) — *Feb 2021 - Jun 2023*

- Graduated with honors
- CONACYT scholarship grantee (2021-2023)
- Advisor: Ruben Santamaria

Instituto Politecnico Nacional — *CDMX, MX*

Bachelor of Science in Physics and Mathematics — *2014 - 2018*

- IPN scholarship grantee (2017)
- Advisor: Juan Ignacio Rodriguez

Publications

1. Acosta García, Bryan Ashley and **Rodriguez Leon, Angel Ivan** and Santamaria Ortiz, Ruben.
   ”Transition-metal doped C_{60} fullerene for temozolomide anticancer drug delivery: a theoretical study”,
   Manuscript under review (Diamond and Related Materials).
   DOI: https://ssrn.com/abstract=5670258

2. **Angel Ivan Rodriguez-Leon**, Cristian Ordóñez, and Ruben Santamaria.
   ”Simulating the helicase enzymatic action on ds-DNA: a first principle molecular dynamic study”,
   ACS Omega 2025, 10, 4, 3627–3639; DOI: 10.1021/acsomega.4c08555

Honors and Awards

*** Fulbright Garcia-Robles Grant**
COMEXUS (2023-current)

*** Opportunity Funds Scholarship**
EducationUSA, Mexico (2023)

*** CONACYT Fellowship**
CONACYT (2021-2023)

Computational Skills

- **Programming & HPC:**
  Python, Fortran, Bash, MPI, OpenMP, CUDA basics, workflow automation and large-scale job management on Linux clusters
- **Molecular Dynamics & Quantum Chemistry:**
  LAMMPS, GROMACS, ASE, MDAnalysis,
  ML interatomic potentials (MACE, NequIP, Allegro, UMA),
  VASP, Quantum ESPRESSO, ORCA, Gaussian, NWChem, TeraChem, Multiwfn
- **Materials Informatics & Cheminformatics:**
  pymatgen, matminer, RDKit
- **Machine Learning & Data Science:**
  PyTorch (basic), JAX (basic), scikit-learn, TensorFlow
- **Visualization & Design:**
  Scientific rendering: VMD, PyMOL, ChimeraX, Ovito, Avogadro, Blender
  Adobe Illustrator, Photoshop, Inkscape, LaTeX
- **Operating Systems & Tools:**
  Git, SLURM, Docker, Jupyter, conda, Linux, macOS, Windows

Teaching Experience

Teaching Assistant, **Chemistry Department, Northwestern University.**
2023–present

- *Advanced General Inorganic Chemistry (CHEM 171)*:
  led recitations for a group of 25 students. Evaluated exams and assignments. Held office hours and provide guidance in problem-solving.
- *Advanced Physical Chemistry Lab (CHEM 182) & Advanced Lab III (CHEM 350-3)*:
  supervised experiments, data analysis, and report writing. Gas phase UV-Vis training.

Pedagogy Training Program and Teaching Assistant, **Instituto de Quimica, UNAM.**
Aug 2023 – Feb 2024

- Completed intensive teacher-training course focused on inclusive pedagogy, gender perspective, and socio-historical contexts in science education
- Graded and tutored the undergraduate course *"Structure of Matter"* (quantum mechanics, bonding models, intermolecular forces topics)

Teaching Assistant – M.Sc. Admission Course (Physics), **Instituto de Fisica, UNAM.**
Sep – Dec 2022

- Designed and graded problem sets for the graduate-level *"Modern Physics"* entrance exam course
- Provided one-on-one and group tutoring in:
  quantum mechanics, statistical physics, particle physics, and atomic/molecular physics.

Work Experience

Specialized Analyst – Basic & Frontier Science, **CONAHCYT (Mexican National Research Council).**
Feb 2024 – Aug 2024

- Oversaw monitoring and completion of >8,000 federally funded basic-science projects
- Designed and implemented a machine-learning + NLP automation platform for project review: integrated databases, multi-criteria search, automated document validation, and internet/PDF-based consistency checks
- Reduced human error and review time by introducing statistical performance metrics and automated verification workflows, significantly increasing efficiency and accuracy of the national project evaluation pipeline

Research Projects

*** Machine-learning interatomic potentials with on-the-fly charge equilibration for mixed ionic–electronic transport in MoS_2 iongels.**
Ongoing project. *Northwestern U*
2025

- Rigorous validation of high-accuracy machine-learning interatomic potentials (MLIPs) for pristine and defected MoS_2/ionic-liquid iongels, enabling DFT-accurate molecular dynamics simulations at long timescales.
- Integrating on-the-fly charge-equilibration schemes with MLIPs to correctly capture dynamic charge redistribution and polarization effects in hybrid ionic/electronic conductors.
- Green–Kubo calculation of electronic conductivity from current autocorrelation functions, capturing the intimate coupling of ionic rearrangements and electronic fluctuations.

*** Ion Doping Mechanisms in Current Spiking of Organic Mixed Ionic-Electronic Conductors.**
Manuscript under preparation. *Northwestern U*
2025

- Collaborated on a large-scale study of electron transport in solution-processed BBL films. My primary contribution was the full DFT-based parametrization of the hopping transport model subsequently used in kinetic Monte Carlo simulations.
- Performed DFT calculations on MD snapshots at different BBL doping levels to generate Gaussian site-energy distributions, reflecting static and dynamic disorder.
- Fitted distance-dependent electronic couplings across 300 independent configurations and provided the final exponential model and cutoff used in KMC simulations.

*** Transition metals doped C60 fullerene as a nanocarrier for temozolomide anticancer drug delivery. A theoretical study.**
Available at SSRN: https://ssrn.com/abstract=5670258.
Oct. 2025

- Studied metal-doped fullerenes to find suitable candidates for drug adsorption and delivery systems in brain cancer treatments.
- Analyzed how the metallofullerenes interact with the temozolomide drug by considering their geometrical, electronic, and energetic properties during adsorption, revealing the significance of doping fullerenes to increase the efficiency of delivery systems.

*** Simulating the helicase enzymatic action on ds-DNA: a first principle molecular dynamic simulation.**
ACS Omega 2025, 10, 4, 3627–3639
Dec. 2024

- Conducted a first principle molecular dynamic simulations on short ds-DNA molecules to analyze their mechanical properties under external forces, providing a way to calculate nucleic acid base separation forces considering specific sequence and stacking interactions along the strand at a quantum mechanical level.
- Developed a computational subroutine for double-helix dissociation that approximates the enzymatic behavior of helicases in cellular processes.

*** Theoretical study of polymers used in fullerene-free organic solar cells: a quantum chemistry computational analysis.**

Bachelor's degree thesis. *IPN*

2019 - 2020

- Conducted a theoretical study on specific molecules used in fullerene free organic solar cells.
- Performed DFT and TD-DFT calculations to analyze factors such as energy gap values, electron transitions, and UV/vis spectra, which suggested the potential of using fullerene-free molecules in the development of highly efficient organic solar cells.

Certificates and Internships

*** Visiting Graduate Student Program**

Visiting Graduate Student Program, NST Division at Argonne National Laboratory *2025-2026*

*** High performance computing graduate program:**

Computer Science and Engineering Program, Instituto de Investigaciones en Matematicas Aplicadas y Sistemas (IIMAS-UNAM) *2023*

*** Molecular Dynamics summer camp:**

11th Molecular Dynamics Summer Workshop, Instituto de Ciencias Físicas (ICF-UNAM) *July 2022*

Presentations and Symposiums

"Simulating the helicase enzymatic action on ds-DNA: a first principle molecular dynamic study". *Houston, Texas*

* Gulf Coast Undergraduate Research Symposium, Chemistry Division, Rice University (Oral Presentation) *21-Oct. 2023*

Morelia, Michoacan

* 66th Mexican Congress of Physics- Mexican Physical Society (Poster). *14-Oct. 2023*

Languages

*** Spanish**

Native

*** German**

Conversational

- ENALLT (UNAM): 3 semesters level A2.1

*** English**

Fluent

- IELTS- level C1 (overall score-8)
- TOEFL- total 107 (R-26, L-28, S-28, W-25)
- GRE: Verbal 158/ Quant. 157 / Writing 3

References available upon request.